6.8.2 Prove Legendre relation $\eta_2(\Lambda)\,w_1 - \eta_1(\Lambda)\,w_2 = 2\pi i$.

Pf: Construct $P = \text{int}\{0, w_1, w_1+w_2, w_2\}$ parallelogram

$$\begin{tikzcd} & \int_0^{w_1}\zeta + \eta_2 w_1 & \\ -\int_0^{w_2}\zeta & & \int_0^{w_2}\zeta - \eta_1 w_2 \\ & \int_0^{w_1}\zeta \arrow[ul] \arrow[ur] & \end{tikzcd}$$

$$\int_{\partial P} \zeta(z)\,dz = \eta_2 w_1 - \eta_1 w_2$$

(~~$\zeta(z_1) = \zeta(z-w_1) + \eta_1$~~).
$\eta_1 = \zeta(z+w_1) - \zeta(z)$.

By residue theorem, $\int_{\partial P} \zeta(z)\,dz = 2\pi i \sum \text{Res}\, \zeta(z)$ in P

$\zeta(z)$ has simple pole at 0 with residue 1

since $\zeta(z) = \frac{1}{z} +$ holomorphic.

Thus $\int_{\partial P} \zeta(z)\,dz = 2\pi i \cdot 1 = 2\pi i$

4.8.3 Actually $\eta_2(\Lambda_\tau) = \frac{(2\pi i)^2}{12}\left(-1 + 24\sum_{m=1}^{\infty}\frac{mq^m}{1-q^m}\right)$. Prove $= G_2(\tau)$.

Pf: $G_2(\tau) = \sum_{(c,d)}{}' \frac{1}{(c\tau+d)^2} = 2\zeta(2) - 8\pi^2\sum_{n=1}^{\infty}\sigma(n)q^n$

$$\eta_2(\Lambda_\tau) = -\frac{(2\pi i)^2}{12} \pm 8\pi^2 \sum_{m=1}^{\infty} m \sum_{l=1}^{\infty} q^{ml}$$

$$= \frac{\pi^2}{3} - 8\pi^2 \sum_{n=1}^{\infty} q^n \sum_{0<l\mid n} \frac{n}{l} = 2\zeta(2) - 8\pi^2\sum_{n=1}^{\infty}\sigma(n)q^n.$$

4.8.4 (a) Verify $F_1^{\bar{v}}$ well-defined

Pf: $F_1^{\bar{v}}(\mathbb{C}/\Lambda, (w_1/N+\Lambda, w_2/N+\Lambda)) = Z_\Lambda\left(\frac{c_v w_1 + d_v w_2}{N}\right) - \frac{c_v\eta_1(\Lambda) + d_v\eta_2(\Lambda)}{N}$.

If $\begin{cases} w_1/N + \Lambda = w_1'/N + \Lambda \\ w_2/N + \Lambda = w_2'/N + \Lambda \end{cases}$ then $\begin{cases} w_1' = w_1 + \alpha N w_1 \\ w_2' = w_2 + \beta N w_2 \end{cases}$

$$\text{RHS} = Z_\Lambda\left(\frac{c_v(w_1+\alpha N w_1) + d_v(w_2+\beta N w_2)}{N}\right) - \frac{c_v\eta_1(\Lambda) + d_v\eta_2(\Lambda)}{N}$$

$$= Z_\Lambda\left(\frac{c_v w_1 + d_v w_2}{N} + c_v\alpha w_1 + d_v\beta w_2\right) - \frac{c_v\eta_1(\Lambda) + d_v\eta_2(\Lambda)}{N}$$

$$= Z_\Lambda\left(\frac{c_v w_1 + d_v w_2}{N}\right) + c_v\alpha\eta_1(\Lambda) + d_v\beta\eta_2(\Lambda) - \frac{c_v\eta_1(\Lambda) + d_v\eta_2(\Lambda)}{N}$$

$$= \cdots \pm \frac{c_v(1-\alpha N)\eta_1(\Lambda) + d_v(1-\beta N)\eta_2(\Lambda)}{N} = \cdots - \frac{c_v\eta_1(\Lambda) + d_v\eta_2(\Lambda)}{N}$$

$(c_v, d_v) \in (\mathbb{Z}/N\mathbb{Z})^2$.

(4.8.4) (b) Verify $F_1^{\bar{v}}$ homogeneous of degree 1 wrt $\Gamma(N)$.

pf: $F_1^{\bar{v}}(\gamma(\mathbb{C}/\Lambda), \gamma(w_1/N+\Lambda, w_2/N+\Lambda))$

$= F_1^{\bar{v}}(\mathbb{C}/\Lambda, (\gamma w_1/N+\Lambda, \gamma w_2/N+\Lambda))$

$= Z_\Lambda\left(\frac{c_v \gamma w_1 + d_v \gamma w_2}{N}\right) - \frac{c_v \eta_1(\Lambda) + d_v \eta_2(\Lambda)}{N}$

$\Lambda' = \gamma\Lambda \Rightarrow \zeta_{\Lambda'}(\gamma z) = \gamma^{-1} \zeta_\Lambda(z)$. $\gamma = (c'\tau + d')$.

Thus $Z_{\Lambda'}\left(\frac{c_v w_1' + d_v w_2'}{N}\right) = (c'\tau + d')^{-1} Z_\Lambda\left(\frac{c_v w_1 + d_v w_2}{N}\right)$

$$\begin{pmatrix} \eta_1(\Lambda') \\ \eta_2(\Lambda') \end{pmatrix} = (c'\tau + d')^{-1} \begin{pmatrix} a & b \\ c' & d' \end{pmatrix} \begin{pmatrix} \eta_1(\Lambda) \\ \eta_2(\Lambda) \end{pmatrix}$$

$$\Rightarrow c\eta_1(\Lambda') + d\eta_2(\Lambda') = (c'\tau + d')^{-1}(c\eta_1(\Lambda) + d\eta_2(\Lambda)).$$